Research Statement

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I am an applied microeconomist with a focus on how environmental exposures and policy changes affect health outcomes and economic wellbeing. A central challenge in this area is that researchers cannot randomly assign pollution, health behaviors, or policy exposure, which makes the causal relationship hard to identify. My work addresses this by using instrumental variables and difference-in-differences designs to produce credible causal estimates from restricted access data. Across my research, I use these methods to answer questions about air pollution, smoking, early childhood outcomes, and reproductive policy. The common thread is a focus on the populations who bear the highest costs of these exposures, and a commitment to getting the identification right so that the findings are reliable enough for policymakers to use.

My job market paper, "Air Pollution and Cardiovascular Mortality: Evidence from Thermal Inversions in the U.S.," estimates the causal effect of fine particulate matter (PM2.5) on cardiovascular mortality, the leading cause of death in the United States. The central empirical challenge is that PM2.5 exposure is not random. Local economic activity affects both pollution and health, satellite-based measures of PM2.5 carry measurement error, and unobserved factors like regional health infrastructure drive both pollution levels and mortality, biasing OLS estimates downward. Most studies using thermal inversions as an instrument rely on a single construction method without testing whether that choice affects instrument validity. Decisions such as pressure-level selection, how an inversion event is defined, and county-level aggregation all affect instrument strength and the plausibility of the exclusion restriction, and I choose my preferred specification based on these diagnostics rather than the largest or most significant effect. This approach delivers a first-stage F-statistic of 463 and a second-stage estimate showing that a one-unit increase in PM2.5 raises cardiovascular mortality by 3.8 percent, a result that holds across multiple robustness checks. The effects are largest for Black, female, elderly, and low-income populations, with direct implications for how policymakers should value and distribute the benefits of air quality regulation.

A second project, "Mental Health and Functional Consequences of Smoking: A Causal Approach Using Misreporting Corrections," examines how smoking affects mental health and daily functioning. Two problems complicate this: smokers differ from nonsmokers in ways that affect health regardless of smoking, and self-reported smoking is prone to underreporting, so measurement error biases even instrumented estimates. I address both by combining a lagged cigarette tax instrument with a bivariate probit model that separately estimates the probability of being a true smoker and the probability of reporting honestly. Preliminary estimates indicate that smoking raises the probability of poor mental health and functional limitations by more than standard estimates suggest, illustrating how much measurement error in health behavior data can distort causal inference.

A third project, "Air Pollution and Early Childhood Outcomes: Evidence from Georgia Childcare Providers" (with Garth Heutel), studies whether pollution keeps preschool-aged children out of childcare, a channel that researchers have largely overlooked relative to birth outcomes or test scores. Using restricted administrative data from the Georgia Department of Early Care and Learning, obtained through a partnership with the Georgia Policy Labs, and thermal inversions as the instrument, we find that a one unit increase in county-week average particulate pollution decreases childcare attendance by about two percentage points, with larger effects for racial minorities and children with working parents. We have posted the paper as an NBER working paper and submitted it to the Journal of Environmental Economics and Management (JEEM).

Outside my dissertation, I am developing a paper titled "The Effect of Targeted Regulation of Abortion Providers (TRAP) Laws on Food Insecurity in the U.S." TRAP laws restrict abortion access through supply-side requirements on providers. Using a staggered difference-in-differences approach over 2010 to 2019, I estimate whether abortion restrictions impose broader financial costs on households beyond direct healthcare access.

My research agenda extends this work in two directions. The first follows directly from my job market paper: many quasi-experimental designs in environmental economics, not just thermal inversions, rest on construction choices that researchers rarely stress-test, and I bring the same diagnostic approach to other instruments and settings where researchers have defaulted to a single specification. The second extends the substantive focus to environmental health domains where medical literature has established associations but no credible causal estimates exist. Economists have studied air pollution more than any other exposure, but exposures such as water contaminants, extreme heat, and wildfire smoke impose large and poorly measured costs on households. One project in this agenda examines how nitrate contamination in drinking water affects thyroid health. Nitrate is the most common chemical contaminant in U.S. drinking water and the medical literature consistently links it to thyroid disease, but no economics paper has produced a causal estimate. Credible estimates here let policymakers trace the full economic cost of water contamination in a way that the medical literature alone cannot. This agenda draws on both my solo-authored work and ongoing collaborations, including with Garth Heutel, and I continue building projects in both formats.